

TED (10) 5058

RegNo:.....

Signature.....

Revision 2010

Model Question paper

SIXTH SEMESTER DIPLOMA EXAMINATION IN ENGINEERING/TECHNOLOGY

BIOPHOTONICS

Maximum Marks:100

MD

Time:3 Hours

PART .A

I. Answer the following questions in one or two sentences each question carries 2 marks

- 1) Mention the characteristics of laser?
- 2) Name any two active material used for laser?
- 3) Define refractive index?
- 4) Give the principle of spectroscopy?
- 5) Give any two therapeutical application of laser?

5x2=10

PART.B

II. Answer any five of the following. Each question carries 6 marks.

1. Describe principle of laser?
2. Describe pumping?
3. Mention the optical properties of tissue?
4. What are the factors effecting the light tissue interaction?
5. Explain optical biosensor?
6. Explain photo sensitizers?
7. Explain flow cytometry?

5x6=30

PART.C

Answer one full question from each unit.

Unit - I

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| III. | a) Explain He-Ne laser with neat diagram? | 12 |
| | b) Mention population inversion? | 3 |

OR

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|-----|---------------------------------------|----|
| IV. | a) Explain any two gas lasers ? | 12 |
| | b) Write note on semiconductor laser? | 3 |

Unit – II

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| V. | a) Explain four photobiological optical properties? | 12 |
| | b) Mention absorption and scattering? | 3 |

OR

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| VI. | a) Explain light tissue interaction in detail? | 12 |
| | b)) Write note on diffused photon density waves ? | 3 |

Unit - III

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| VII. | a) Describe the principle of bio imaging ? | 10 |
| | b) Explain optical sensor? | 5 |

OR

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| VIII. | a) Explain optical coherence tomography? | 9 |
| | b) Write note on in vivo imaging? | 6 |

Unit - IV

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|-----|--|---|
| IX. | a) Give surgical applications of laser? | 8 |
| | b) Describe two techniques of laser therapy? | 7 |

OR

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| X. | a) Describe any three application of laser in dermatology? | 15 |
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